PROJECT AQUAFLUSH

Arduino-Based Water-Flow Monitoring Prototype for Small Urban Drainage Systems

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Document updated October 2026 from the May 2025 presentation and recovered sketch. No new experiments were conducted for this revision.

1. Abstract

Project AquaFlush is a laboratory-scale drainage exhibit motivated by flooding concerns in Barangay Tibanga, Iligan City. The demonstrable Arduino firmware reads pulses from a water-flow sensor, estimates flow in liters per hour, and displays the measurement locally through a 16×2 I²C LCD and serial output. The project also included a miniature roadway and drainage model with additional pump and relay hardware. The presentation proposed automated clog management and IoT-based information sharing; however, those functions are not implemented in the sole Arduino sketch the group reports using. This report documents the implemented monitoring function separately from proposed extensions.

2. Problem Background and Objectives

The May 2025 group presentation highlighted recurring flooding near MSU-IIT and attributed drainage concerns to trash, debris, and constrained drainage systems. The prototype was presented as a small-scale educational demonstration of flow monitoring.

Documented objectives: (1) acquire pulses from a water-flow sensor using an Arduino Uno; (2) estimate and display water-flow readings; (3) integrate electronics into a miniature drainage model; and (4) discuss further sensing and automation needed for practical clog detection.

3. Components and Tools

Microcontroller	**Arduino Uno / ATmega328P**
Input	Pulse-output water-flow sensor (exact model unverified)
Local display	16×2 I²C LCD
Additional exhibit hardware	5 V relay and water pump, shown/listed but not controlled in recovered sketch
Development	Arduino IDE; Arduino C/C++; Wire and LiquidCrystal_I2C libraries
Construction	Jumper wiring, mounted electronics, miniature drainage/roadway model

4. Firmware and Measurement Method

The recovered sketch attaches an interrupt handler to the flow sensor on Arduino Uno digital pin D2 (external interrupt 0). A counter increments on each rising pulse. Roughly once per second, the accumulated pulses are converted to an estimated water-flow value in L/hour and printed on the LCD and Serial Monitor (9600 baud).

The sketch uses the expression l_hour = (flow_frequency × 60) / 7.5. The divisor 7.5 is an assumed sensor calibration factor, not a measured calibration specific to the exhibited hardware. The program does not implement automated relay switching, pump timing, internet transfer, a clog threshold, or a validated alarm.

5. Hardware and Prototype Construction

As shown in the original presentation, the group built a tabletop miniature drainage environment and mounted the LCD and electronic assemblies around the model. The wiring photograph shows electronics and a relay module; the pump and water container are also visible beneath the display structure. Exact physical wiring and component ratings should be re-verified before reuse.

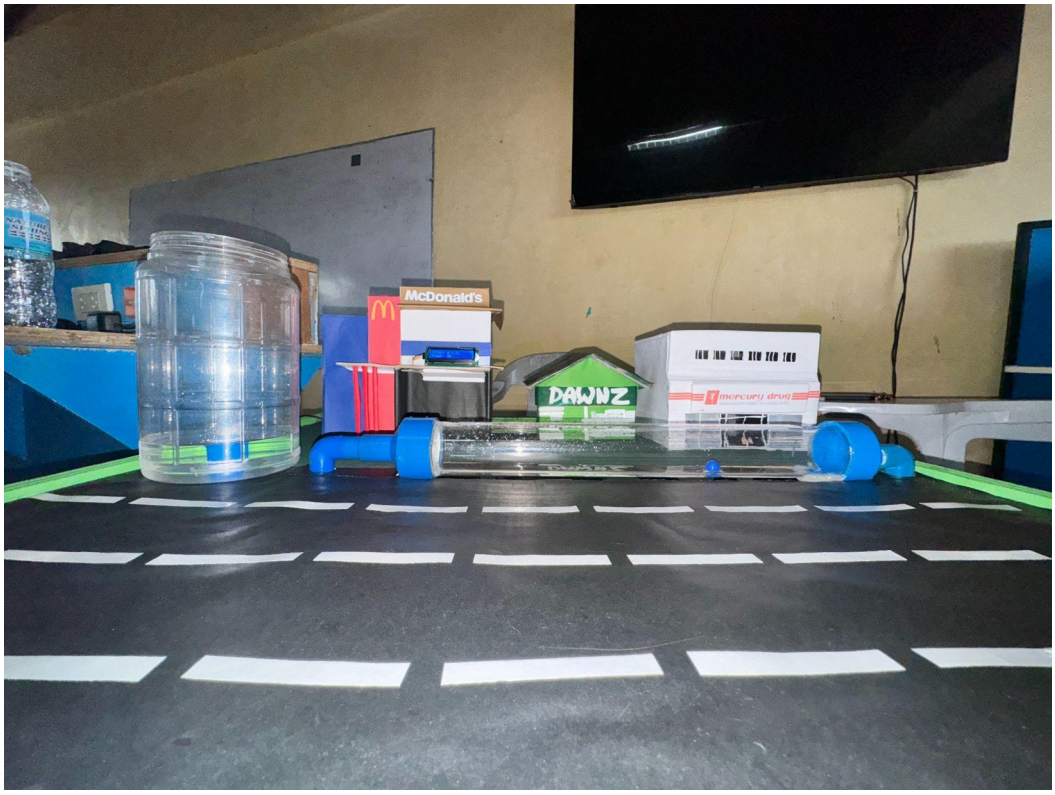


Figure 1. Miniature roadside drainage model, photographed for the May 2025 presentation.

6. Project Documentation

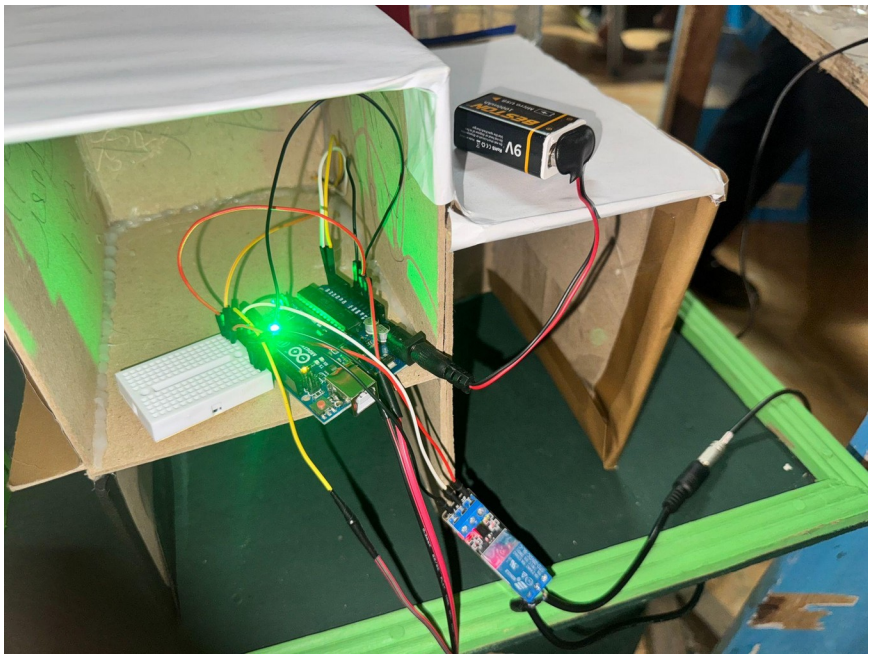


Figure 2. Electronics and wiring arrangement inside the prototype.

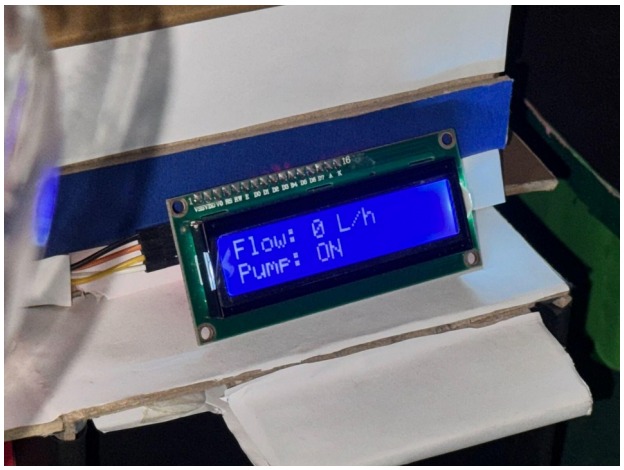


Figure 3. LCD during the demonstration (photographic evidence).

Note: The photo shows a pump-status message, but no corresponding LCD pump-status logic appears in the recovered sole firmware. The photograph therefore does not establish that the Arduino sketch controlled a pump.

7. Observations, Results, and Limitations

The supplied presentation documents a constructed prototype, water-flow display hardware, firmware screenshots, and demonstration photographs. It does not provide time-series measurements, controlled clog/non-clog trials, measured accuracy, pump activation logs, or verified internet communication. Consequently, quantitative performance and automatic clog detection cannot be claimed based on these materials.

Key technical limitations are the unverified sensor calibration, lack of clogging classification, absence of pump/relay control logic in the recovered code, and lack of online IoT telemetry. Actual drainage obstruction cannot be uniquely inferred from a low-flow value without accounting for source flow and hydraulic conditions.

8. Recommendations

Following the project's original recommendations, future work should improve hydrodynamic design, integrate additional sensors, and study other root causes of flooding. Before deploying automated actions, teams should validate clog detection against controlled test cases and implement safe, electrically isolated actuator control. Connectivity and an LGU monitoring dashboard would require separate implementation and testing.